

1. There are 3 main cases: (1) X caused Y , (2) Y caused X , and (3) X and Y have certain common factor C .

For example, the faster the windmills are observed to rotate, the more wind is observed. This does not mean that wind is caused by the rotation of windmills. It is rather the opposite: the rotation of windmills is caused by wind. If we want to construct a linear regression model for this example, then the dependent variable should be the speed of windmill, and the explanatory variable should be the speed of wind:

$$S_{windmill} = \beta_0 + \beta_1 S_{wind} + \epsilon, \quad (1)$$

where $S_{windmill}$ denotes the speed of windmill, S_{wind} denotes the speed of wind, and ϵ denotes the other factors that may affect the speed of windmill such as mechanical factors. Again, here we are interested in the parameter $\beta_1 = \Delta S_{windmill} / \Delta S_{wind}$: given one unit change in the speed of wind, what is the corresponding change in the speed of windmill.

For the case with certain common factor C , consider the following example: as ice cream sales increase, the rate of drowning deaths increases. Ice cream sales and death rate do not have causal relationship, but they have the common factor “temperature”. Here, it does not make sense to construct a linear regression model for ice cream sales and rate of drowning death. But we can consider the following linear regression model:

$$Sales = \beta_0 + \beta_1 Temperature + \epsilon, \quad (2)$$

where $Sales$ denotes the ice cream sales, and we can study the effect of temperature on the sales. Similarly, the following model also makes sense:

$$Death = \beta_0 + \beta_1 Temperature + \epsilon, \quad (3)$$

where $Death$ denotes the rate of drowning death.

Similarly, if we observe that the consumption of masks and the hours of online working are positively correlated with each other, this is not because they have causal relationship. Instead, there exists a common factor, which is the Covid.

2. For the example of class size and student performance, the linear regression model can be written as

$$\text{Test Score} = \beta_0 + \beta_1 \text{STR} + u,$$

where “Test Score” denotes the school district-level average test score, “STR” denotes the

school district-level average student-to-teacher ratio, and u denotes the error term. β_0 and β_1 are the parameters of the model. β_0 is the intercept parameter, and β_1 is the slope parameter. Here, we are more interested in β_1 because it represents the change in “Test Score” given one unit change in “STR”. The error term u represents the other factors that affect “Test Score”.

3. (a) The term u_i represents the factors other than the nap time that influence the participant’s performance on the short test, including inherent cognitive ability, aptitude, etc. Different students have different u_i because some may have better (worse) inherent ability than others.
- (b) The conditional mean can be considered a group mean. Therefore, $E[u_i|X_i]$ means the average value of the other factors that influence the participant’s performance among the group of participants who had certain amount of nap time.
For instance, $E[u_i|X_i = 0]$ means the average value of the other factors among the group of participants who had 60 minutes of nap time. Similarly, $E[u_i|X_i = 1]$ means the average value of the other factors among the group of participants who had 75 minutes of nap time.
On the other hand, the unconditional mean $E[u_i]$ mean the average value of the other factors among all participants.
- (c) Yes, we expect that in this experiment, $E[u_i|X_i] = E[u_i]$. This is because the random assignment of participants ensures that u_i is totally independent from X_i . Then, by the property of conditional expectation, given the independence between u_i and X_i , we have $E[u_i|X_i] = E[u_i]$, i.e., the conditional expectation is equal to the unconditional expectation.
4. *For the Review of Probability, you may refer to Chapter 2 of “Introduction to Econometrics”. The following topics are highly relevant to our course: **the expected value of a random variable, the standard deviation and variance, conditional expectation, independence, covariance and correlation, the mean and variance of sums of random variables.**

- (a) Given $Y = X^2 + Z$, we have

$$E[Y|X] = E[X^2 + Z|X] = E[X^2|X] + E[Z|X] = X^2 + E[Z] = X^2,$$

where the second equality follows from the fact that **we can always interchange the order of summation and expectation**, the third equality follows from the property of conditional expectation and the independence between Z and X , and the last equality follows from $E[Z] = 0$ as Z is a standard normal random variable.

(Note: In general, $E[g(X)|X] = g(X)$, where $g(X)$ is a certain function of X such as X^2 , X^3 , etc. Intuitively, given the information (value) of X , we will know the value

of X^2 , e.g., if $X = 1$, then $X^2 = 1$; if $X = 2$, then $X^2 = 4$. Therefore, given X , there will be no more randomness in X^2 , so its expected value is equal to itself. This is the reason why $E[X^2|X] = X^2$.)

(b)

$$E[Y] = E[X^2 + Z] = E[X^2] + E[Z] = 1 + 0 = 1,$$

where the third equality follows from $\text{Var}[X] = E[X^2] - (E[X])^2$ and X is a standard normal random variable so that $E[X] = 0$ and $\text{Var}[X] = 1$.

(c)

$$E[XY] = E[X(X^2 + Z)] = E[X^3 + XZ] = E[X^3] + E[XZ] = 0 + E[X]E[Z] = 0 + 0 = 0,$$

where the fourth equality follows from $E[X^3] = 0$ as X is a standard normal random variable (in fact the odd moments of a standard normal random variable are all zero), and $E[XZ] = E[X]E[Z]$ as X and Z are independent from each other. Note that **we can interchange the order of expectation and multiplication if the two random variables are independent**. This is very different from interchanging the order of expectation and summation, where the independence condition is not needed.

(Note: Given that X is standard normal, it is a continuous random variable. Therefore, $E[X^3] = \int_{-\infty}^{+\infty} x^3 f(x) dx = \int_{-\infty}^0 x^3 f(x) dx + \int_0^{+\infty} x^3 f(x) dx$, where $f(x)$ denotes the PDF of a standard normal random variable. Based on the property of the standard normal distribution, $f(x)$ is symmetric around zero. Therefore, $\int_{-\infty}^0 x^3 f(x) dx = -\int_0^{+\infty} x^3 f(x) dx$ so that $E[X^3] = 0$. By the same reasoning, $E[X^k] = 0$ if k is an odd number.)

5. Let H_0 and H_1 denote the null hypothesis and the alternative hypothesis, respectively.

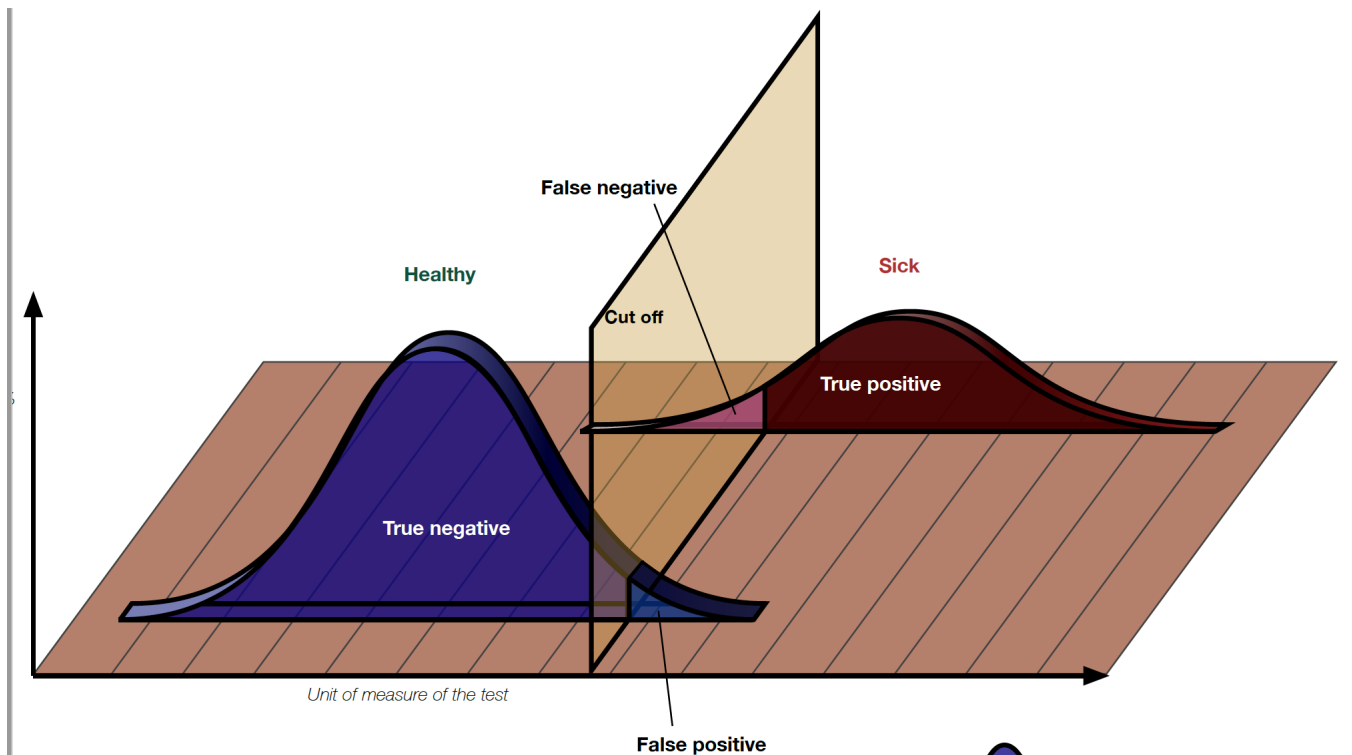
Type I error: reject H_0 when H_0 is true.

Type II error: accept H_0 when H_1 is true.

There exists a trade-off between Type I and Type II errors in the sense that in general we cannot minimize the two types of errors at the same time.

We take the covid-19 antibody test (ART test) as an example. If we want to minimize the Type I error (false positive) of the antibody test, we may increase the **threshold value (critical value)** of the antibody level used in the test. Suppose that the original threshold is 10 units, and the test claims that a person is infected if the actual antibody level in his or her body is higher than 10 units (i.e., the test will reject H_0 if his or her antibody level is higher than 10 units).

If we change the **threshold from 10 units to 100 units**, then the **Type I error will decrease** because now it is much more difficult to reject H_0 (now the test will reject H_0 only when the person's antibody level is higher than 100 units).



However, the **Type II error will increase** at the same time because there will be much more false negative cases: since it is now very difficult to reject H_0 , for many people the test will claim that they are not infected even if they are actually infected (i.e., now the test does not have detecting power).

Similarly, if we change the **threshold from 10 units to 1 unit**, then the **Type II error will decrease** because by doing so it will be very easy to reject H_0 (now the test becomes very sensitive as it will reject H_0 as long as the person's antibody level is higher than 1 unit). On the other hand, the **Type I error will increase** because there will be much more false positive cases.

The Type I and II errors (false positive and false negative) of a Covid antibody test are illustrated in the following figure. In the figure, the blue distribution represents the distribution of the antibody levels observed from uninfected people, while the brown distribution represents the distribution of the antibody levels observed from infected people. Notice that if we move the threshold toward right (i.e., increase the threshold value/critical value), the Type I error will decrease while the Type II error will increase. By contrast, if we move the threshold toward left (i.e., reduce the threshold value/critical value), the Type I error will increase while the Type II error will decrease. Therefore, there exists a trade-off.